

Masks that calibrate themselves

Can the pixel masks of Skipper-CCD analyses do their job without being tuned to the particular sensor they run on?

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Short answer, from simulations of three different sensors (5 independent seeds). Of the 28 cases in which a mask tuned by hand on one sensor was moved unchanged to another, 5 were *worse than applying no mask at all* in every seed, and 2 made no difference. The clearest was low-energy clusters on the surface laboratory sensor (tuned on deep underground): a median 0.12 of the oracle, where doing nothing scores 1.00.

The same masks written as procedures that calibrate themselves were worse than no mask in every seed in 1 of 14 cases (low-energy clusters on the surface laboratory sensor (adaptive)), and better than no mask in every seed in 12. Measured against a mask tuned with truth on the same sensor they reach a median 0.98, with a worst case of 0.38 as a median over seeds (charge-transfer trails on the surface laboratory sensor (adaptive); 0.34 in its worst seed). In 8 cases a transplanted mask beat the adaptive one in every seed: self-calibration avoids the large failures but is not free.

On the three sensors with their defects switched off, the adaptive masks fired no more often than α allows (and apparently less often; 50 runs cannot tell). On the surface sensor, though, the adaptive low-energy-cluster mask keeps only 0.85–0.92 of the signal, 0.85 even in a seed with no low-energy clusters at all: it fires on charge-transfer trail electrons, a false positive that the defect-free test cannot see because it switches every defect off at once, which is why the defects were also switched on one at a time. Switching the defects on one at a time instead of all off at once shows it directly: of the 60 combinations in which the defect present is not the mask's own, 7 fire on it, the worst in 1.00 of trials while masking 0.121 of the image. On a real sensor that nothing here was tuned for, the four exposures of the public SENSEI release, the hot-column procedure ends up flagging the 7 loudest columns of the longest exposure and nothing else, all of them inside the mask the collaboration published; it masks 0.0007–0.0094 of the image where that published mask takes 0.042–0.130.

On the 2 seeds run once after the code was frozen: 4 of 26 transplanted cases worse than no mask in every seed, 0 of 13 adaptive ones, and a median adaptive 0.98 of the oracle. The adaptive case that is worse than no mask in every development seed (low-energy clusters on the surface laboratory sensor (adaptive)) has fewer than 20 target events in the held-out seeds, so it is not counted there, although there too its figure of merit is below no mask (0.96, 0.92 of no mask).

Counted pairwise: a case is harmful only if the transplanted mask scores more than 1 % below no mask in every seed with at least 20 target events. The rule pre-registered before the re-run compared medians without a margin and gives 6 harmful transplants and 1 harmful adaptive cases; it was replaced after an independent review showed that medians differing in the third decimal were being counted as harm.

The question

A Skipper-CCD counts single electrons in millions of pixels. Before any physics is extracted, analyses discard pixels with masks, each aimed at one problem: charge left behind during transfer, hits in the serial register, hot columns and pixels, photons emitted around high-energy tracks (the halo), clusters of low-energy events, and muons. Every mask has sizes and thresholds — a radius, a trail length, a rate cut — chosen by hand for one sensor, readout and site.

The working hypothesis: what transfers between sensors is not the numbers but the *procedure* used to choose them. So each mask was written twice: a **fixed** form with hand-set constants, and an **adaptive** form that measures its own sizes on the images it is applied to and sets its threshold by a false-positive rate ($\alpha = 0.01$, corrected for the number of tests) or by measurable physical properties of the sensor.

Mask	Fixed form	Adaptive form
Hot columns & pixels	column rate above $k \times$ median	Poisson tail of each column's count against its own exposure, Bonferroni over columns, single columns before groups
Charge-transfer trails	L pixels after every bright pixel	downstream vs. upstream excess of single electrons around the same bright pixel (binomial); symmetric sources cancel
Halo	disc of radius R around bright pixels	each annulus against everything outside it (conditional binomial); no far field needed
Serial-register hits	rows with n charged pixels in a window of w	densest window of each row against the image's own occupancy, at several widths
Low-energy clusters	disc around events with $\geq m$ neighbours	neighbours within r against a Poisson count for the local valid area, at several radii
Muon tracks	charge and length cuts	charge of a minimum-ionising track for the measured length and this sensor's thickness; straight, piled-up or cut by the border

How it was tested

A simulator that remembers where every electron came from

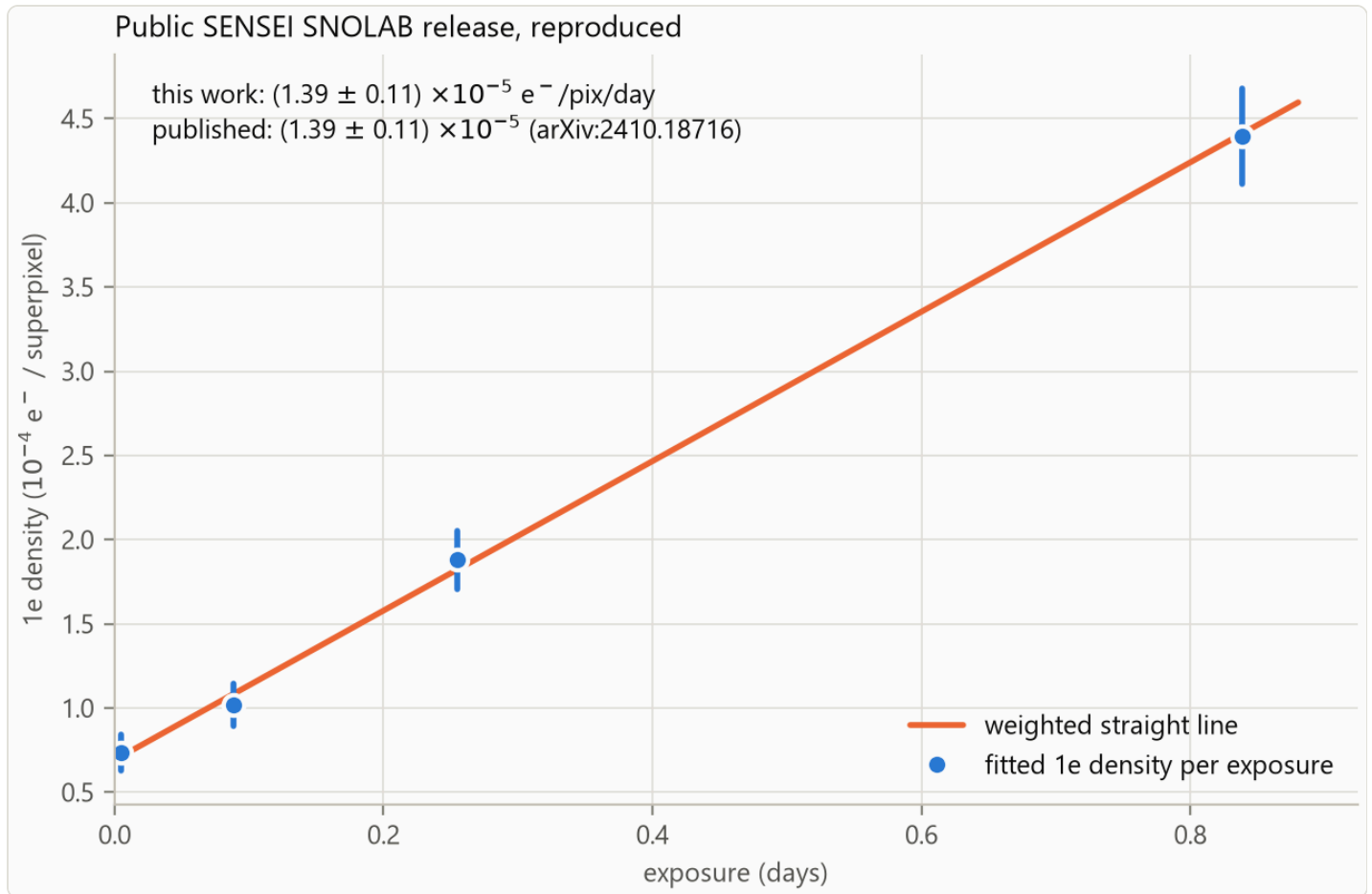
Each source of charge fills its own map, so a mask can be scored against exactly the events it should remove. Three sensors differ the way real deployments do. Their geometry, noise, dark rates, background rates and muon fluxes come from public measurements (SENSEI at SNOLAB and MINOS, Oscura sensors at the surface, SNO, PDG); defect populations are not published in a transferable form and are scenario values chosen to differ between sensors.

Sensor	Pixels	Thickness (μm)	Noise (e^-)	Exposure	Dark rate ($\text{e}^-/\text{pix}/\text{day}$)	Muons (cm^{-2} day^{-1})	High-E (dru)	Halo μm / trail px / hot cols
Deep underground	3072 × 512	665	0.14	20 h	1.4×10^{-5}	2.9e-05	50	100 / 30 / 4
Shallow underground	3072 × 443	675	0.14	20 h	1.6×10^{-4}	6.9	3.37e+03	150 / 60 / 10
Surface laboratory	1278 × 1058	725	0.17	1 h	3.0×10^{-2}	1.4e+03	3e+04	250 / 100 / 25

Three versions of every mask were compared on an independent test stack of each sensor: the **oracle** (fixed form tuned with the simulator's truth on that sensor), the **transplant** (the oracle constants of a different sensor) and the **adaptive** form, calibrated without truth. The figure of merit is $S/\sqrt{S+B}$: surviving injected signal against surviving target background.

A check against real images

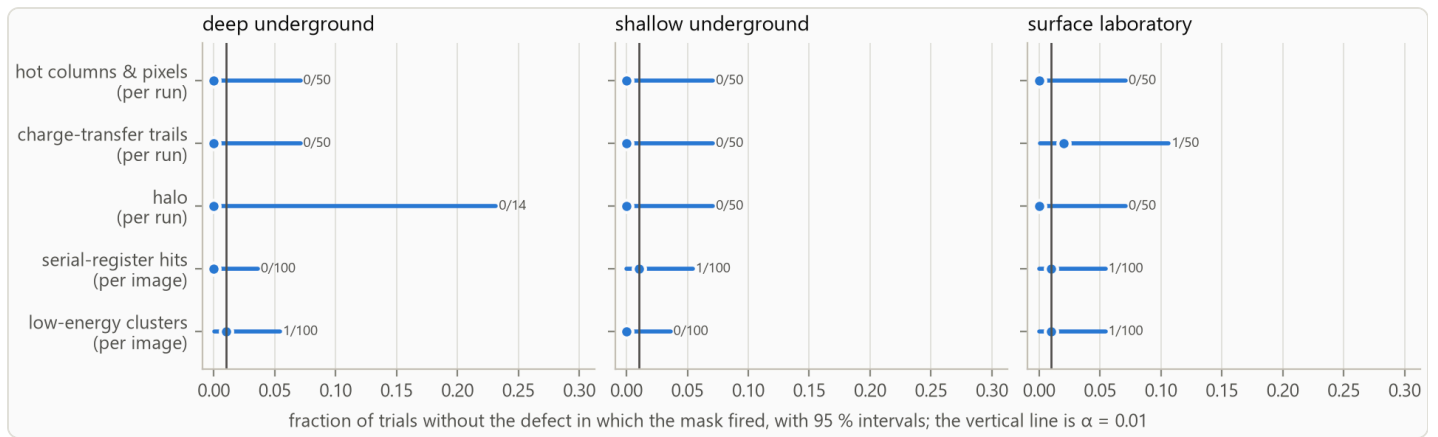
Before relying on anything, the public SENSEI SNOLAB data release was read and its single-electron rate reproduced: $1.39 \times 10^{-5} \pm 1.1 \times 10^{-6} \text{ e}^-/\text{pix}/\text{day}$ against the published $1.39 \times 10^{-5} \pm 1.1 \times 10^{-6}$ (pull -0.02).



Single-electron density per exposure in the public SENSEI SNOLAB release, with the weighted straight line. The slope is the rate.

Result 1 · How often the adaptive masks fire when there is nothing to find

50 independent runs of each of the three sensors with every defect the masks look for switched off, keeping what a real sensor cannot switch off: dark current, spurious charge, the injected signal, muon tracks and high-energy deposits. The stack-calibrated masks decide once per run of 2 images; the others decide per image. The test is two-sided, so a mask that is far *more* conservative than α is visible as well.



Clopper–Pearson 95 % intervals; the vertical line is $\alpha = 0.01$. Intervals are wide because 50 runs cannot resolve 0.01 from 0.03.

Sensor	Adaptive mask	Unit	Fired / trials	Rate	95 % interval	Against α
Deep underground	Hot columns	per run	0 / 50	0.000	[0.000, 0.071]	contains α
Deep underground	Charge-transfer trails	per run	0 / 50	0.000	[0.000, 0.071]	contains α
Deep underground	Halo	per run	0 / 14	0.000	[0.000, 0.232]	contains α
Deep underground	Serial-register hits	per image	0 / 100	0.000	[0.000, 0.036]	contains α
Deep underground	Low-energy clusters	per image	1 / 100	0.010	[0.000, 0.054]	contains α
Shallow underground	Hot columns	per run	0 / 50	0.000	[0.000, 0.071]	contains α
Shallow underground	Charge-transfer trails	per run	0 / 50	0.000	[0.000, 0.071]	contains α
Shallow underground	Halo	per run	0 / 50	0.000	[0.000, 0.071]	contains α
Shallow underground	Serial-register hits	per image	1 / 100	0.010	[0.000, 0.054]	contains α
Shallow underground	Low-energy clusters	per image	0 / 100	0.000	[0.000, 0.036]	contains α
Surface laboratory	Hot columns	per run	0 / 50	0.000	[0.000, 0.071]	contains α
Surface laboratory	Charge-transfer trails	per run	1 / 50	0.020	[0.001, 0.106]	contains α
Surface laboratory	Halo	per run	0 / 50	0.000	[0.000, 0.071]	contains α
Surface laboratory	Serial-register hits	per image	1 / 100	0.010	[0.000, 0.054]	contains α

Sensor	Adaptive mask	Unit	Fired / trials	Rate	95 % interval	Against α
Surface laboratory	Low-energy clusters	per image	1 / 100	0.010	[0.000, 0.054]	contains α

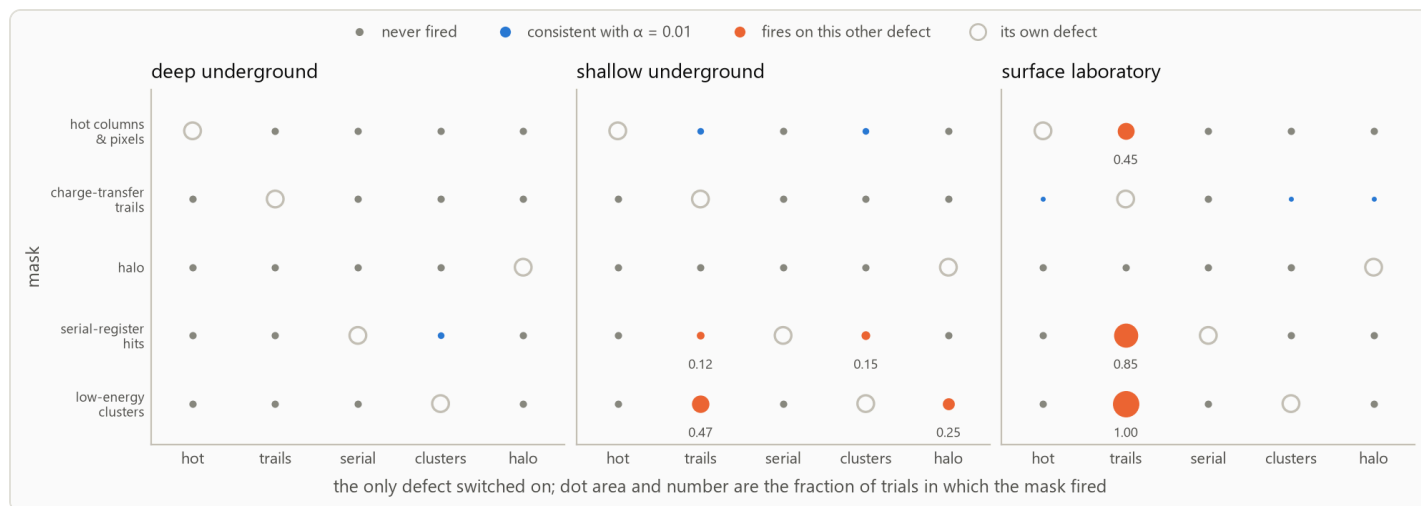
The muon mask is not in that table, and could not be: the five above are switched off by setting a defect rate to zero, while tracks are what a real sensor cannot switch off, and the halo is generated from their charge. Asked separately, on sensors with the tracks and the high-energy background switched off as well, it fired in 0 of 600 images (all three sensors, 200 images each), an upper limit of 0.018 at 95 %. Its threshold is physical rather than a tail probability — a cluster long enough, straight enough and carrying about the charge a minimum-ionising particle leaves crossing this sensor — so there is no α for it to match; what is measured is whether dark current can imitate a track, and at these exposures it cannot.

Result 2 · What a mask fires on when the defect present is not its own

The defect-free test above switches every defect off together, so a mask that fires on *another* mask's defect passes it unseen. Here each sensor is simulated with one defect at its preset value and the other four at zero, 15 configurations in all, 20 runs each, with the same chain of adaptive calibrations and the same things a real sensor cannot switch off. A mask that fires when its own defect is absent and another one is present is firing on the wrong thing.

The same question is asked of 60 mask-and-defect cells, so the bound a cell has to clear is taken at 99.92 % confidence rather than 95 %, which controls a 5 % chance of one false call over the whole grid. The hot-column row is compared with 0.02 rather than α , because it fires when either of two α -level procedures does. Two caveats the design cannot remove: the 2 images of a run share the calibrations made on that run's stack, so for the two per-image masks they are not independent trials and the intervals are narrower than they should be; and a cell marked below as sitting at its detection floor fired the fewest times that could produce this verdict at all.

Of the 60 combinations in which the defect present is not the one the mask looks for, 7 fire more often than α allows and 53 are consistent with it. The largest is the low-energy clusters mask on the surface laboratory sensor when the only defect present is charge-transfer trails: it fires in 1.00 of trials and masks a median 0.121 of the image.



Dot area and number are the fraction of trials in which the mask fired; orange marks the cells whose lower bound, at the 99.92 % confidence corrected for the 60 cells tested, lies above the rate that mask should not exceed. Grey dots never fired; hollow circles are the diagonal, where the mask meets its own defect.

Sensor	Only defect present	Mask that fired	Unit	Fired / trials	Rate	95 % interval	Median fraction masked	Against α
Shallow underground	Charge-transfer trails	Serial-register hits	per image	5 / 40	0.12	[0.042, 0.268]	0.000	fires on it (at the detection floor)
Shallow underground	Charge-transfer trails	Low-energy clusters	per image	19 / 40	0.47	[0.315, 0.639]	0.000	fires on it
Shallow underground	Low-energy clusters	Serial-register hits	per image	6 / 40	0.15	[0.057, 0.298]	0.000	fires on it
Shallow underground	Halo	Low-energy clusters	per image	10 / 40	0.25	[0.127, 0.412]	0.000	fires on it
Surface laboratory	Charge-transfer trails	Hot columns	per run	9 / 20	0.45	[0.231, 0.685]	0.000	fires on it
Surface laboratory	Charge-transfer trails	Serial-register hits	per image	34 / 40	0.85	[0.702, 0.943]	0.001	fires on it
Surface laboratory	Charge-transfer trails	Low-energy clusters	per image	40 / 40	1.00	[0.912, 1.000]	0.121	fires on it

Result 3 · Transplanted constants can do harm; self-calibration avoids the large failures, at a cost

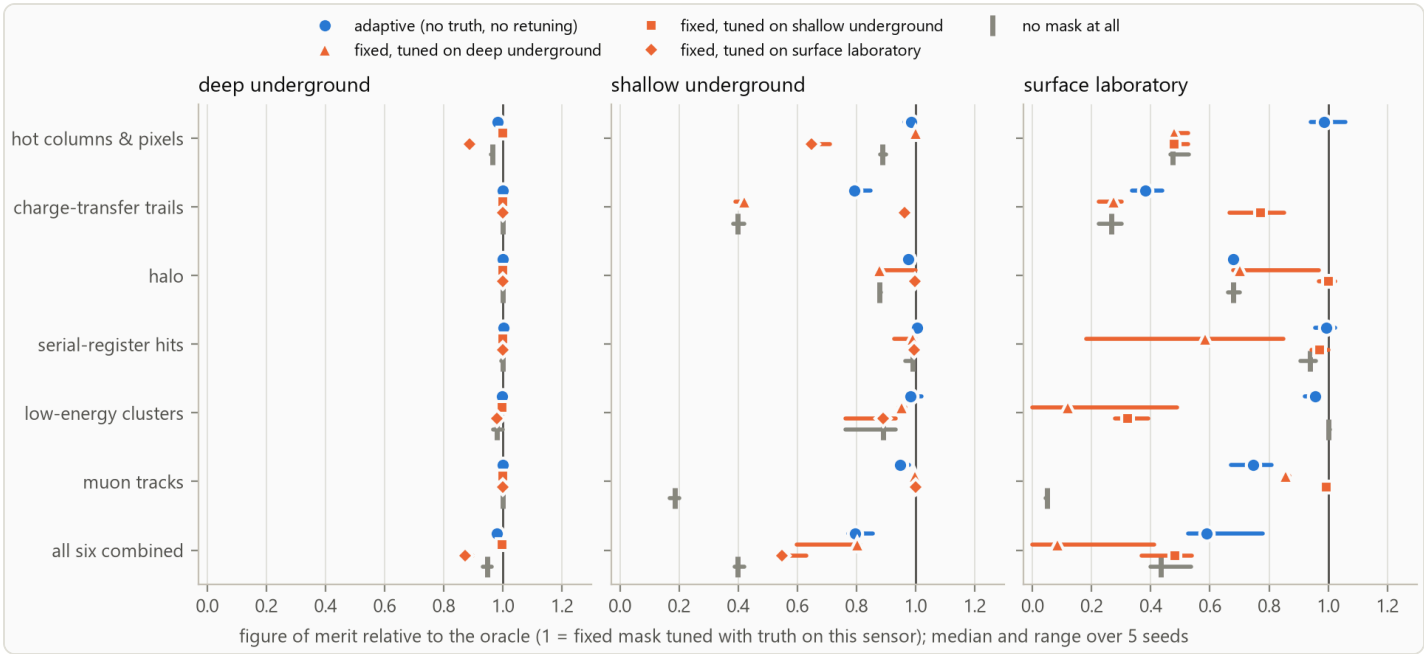


Figure of merit relative to the oracle tuned with truth on the same sensor (vertical line). Markers are medians over 5 seeds, whiskers the range. The deep-underground sensor has few target events, so every version sits near 1 there.

Evaluated on	Mask	Target events	No mask	Adaptive	Fixed, tuned on deep	Fixed, tuned on shallow	Fixed, tuned on surface
Deep underground	Hot columns & pixels	59	0.97 (0.96–0.97)	0.98 (0.98–0.99)	oracle = 1	1.00 (1.00–1.00)	0.89 (0.88–0.90)
Deep underground	Charge-transfer trails	0 too few to compare	1.00 (1.00–1.00)	1.00 (1.00–1.00)	oracle = 1	1.00 (1.00–1.00)	1.00 (1.00–1.00)
Deep underground	Halo	0 too few to compare	1.00 (1.00–1.00)	1.00 (1.00–1.00)	oracle = 1	1.00 (1.00–1.00)	1.00 (1.00–1.00)
Deep underground	Serial-register hits	3 too few to compare	1.00 (0.99–1.00)	1.00 (1.00–1.00)	oracle = 1	1.00 (1.00–1.00)	1.00 (1.00–1.00)
Deep underground	Low-energy clusters	25	0.98 (0.97–1.00)	1.00 (0.99–1.00)	oracle = 1	1.00 (0.99–1.00)	0.98 (0.97–1.00)
Deep underground	Muon tracks	0 too few to compare	1.00 (1.00–1.00)	1.00 (1.00–1.00)	oracle = 1	1.00 (1.00–1.00)	1.00 (1.00–1.00)
Deep underground	All six combined	81	0.95 (0.93–0.96)	0.98 (0.98–0.99)	oracle = 1	1.00 (0.99–1.00)	0.87 (0.86–0.90)
Shallow underground	Hot columns & pixels	176	0.89 (0.88–0.90)	0.98 (0.96–1.00)	1.00 (1.00–1.00)	oracle = 1	0.65 (0.63–0.71)
Shallow underground	Charge-transfer trails	3237	0.40 (0.38–0.42)	0.79 (0.78–0.85)	0.42 (0.39–0.42)	oracle = 1	0.96 (0.95–0.97)
Shallow underground	Halo	211	0.88 (0.88–0.88)	0.97 (0.97–0.99)	0.88 (0.88–1.00)	oracle = 1	1.00 (0.99–1.00)

Evaluated on	Mask	Target events	No mask	Adaptive	Fixed, tuned on deep	Fixed, tuned on shallow	Fixed, tuned on surface
Shallow underground	Serial-register hits	24	0.99 (0.96–0.99)	1.01 (1.00–1.01)	0.99 (0.93–1.00)	oracle = 1	1.00 (0.99–1.01)
Shallow underground	Low-energy clusters	176	0.89 (0.76–0.93)	0.98 (0.96–1.02)	0.95 (0.94–0.97)	oracle = 1	0.89 (0.76–0.93)
Shallow underground	Muon tracks	14116	0.19 (0.17–0.20)	0.95 (0.94–0.98)	1.00 (0.99–1.00)	oracle = 1	1.00 (1.00–1.00)
Shallow underground	All six combined	4093	0.40 (0.39–0.42)	0.80 (0.77–0.86)	0.80 (0.60–0.81)	oracle = 1	0.55 (0.54–0.63)
Surface laboratory	Hot columns & pixels	1657	0.47 (0.47–0.53)	0.99 (0.94–1.06)	0.48 (0.47–0.53)	0.48 (0.47–0.53)	oracle = 1
Surface laboratory	Charge-transfer trails	66249	0.27 (0.22–0.30)	0.38 (0.34–0.44)	0.27 (0.22–0.30)	0.77 (0.67–0.85)	oracle = 1
Surface laboratory	Halo	2049	0.68 (0.66–0.70)	0.68 (0.66–0.70)	0.70 (0.68–0.97)	1.00 (0.97–1.02)	oracle = 1
Surface laboratory	Serial-register hits	76	0.94 (0.91–0.96)	0.99 (0.96–1.02)	0.59 (0.18–0.85)	0.97 (0.94–1.00)	oracle = 1
Surface laboratory	Low-energy clusters	41	1.00 (1.00–1.01)	0.96 (0.92–0.97)	0.12 (0.00–0.49)	0.32 (0.28–0.39)	oracle = 1
Surface laboratory	Muon tracks	175494	0.05 (0.04–0.05)	0.75 (0.67–0.81)	0.86 (0.85–0.87)	0.99 (0.99–1.00)	oracle = 1

Evaluated on	Mask	Target events	No mask	Adaptive	Fixed, tuned on deep	Fixed, tuned on shallow	Fixed, tuned on surface
Surface laboratory	All six combined	70035	0.43 (0.40–0.54)	0.59 (0.53–0.78)	0.08 (0.00–0.41)	0.48 (0.37–0.54)	oracle = 1

Figure of merit relative to the oracle of the sensor in the first column; median, with the range over seeds in brackets.

Result 4 · The same procedures on a real sensor

Everything above is simulated. Here the adaptive masks run unchanged on the 4 exposures of the public SENSEI SNOLAB release: a fourth sensor, real, with a geometry none of the presets has (3072×16 active superpixels, each binning 32 physical rows, so a step along a column is 32 times a step along a row). Nothing about it was tuned, and the release publishes its own mask, so where the two overlap they can be compared. The pairing of one of our masks with a bit of that mask follows the hypothesis tested against the release's own geometry earlier on this page.

Given only the images, the estimator measures a readout noise of 0.141–0.142 e, where the independent fit of the rate reproduction above, on the same files, gives 0.1413–0.1420 e, and a single-electron density that grows with exposure.

What the hot-column procedure finds. It flags 2, 4, 6, 7 columns as the exposure grows, and the sets are nested: each one keeps the previous columns and adds the next loudest. At the longest exposure the 7 columns with the highest rate of charged pixels are exactly the ones it flags, and every one of them lies inside the release's own bad-column mask; the next column down (1659, at 0.019 charged pixels per pixel) is in the release's mask and is not flagged, which is the conservative direction. Each flagged column carries 102 to 1830 times the charged-pixel rate of the columns left alone.

The two directions of agreement say different things, so both are here. Of what we flag, the release also flags 0.99–1.00: we stay inside their mask. Of what they flag, we flag 0.027–0.095: they mask far more than we do, 0.042 to 0.130 of the image against our 0.0007 to 0.0094. The first number is the easy direction — a procedure that flagged one true column would score 1.00 — and it rests on 2 to 7 column decisions, not on the hundreds of pixels it is counted over.

Exposure	Images	Trigger pixels	Measured noise (e)	1e density (e/pix)	Trail lengths h / v	Halo radius	Hot columns / pixels	We mask (union)	The release masks
0 h	14	0	0.142	1.34×10^{-4}	0 / 0	0	2 / 8	0.0007	0.042
2 h	14	70	0.142	1.72×10^{-4}	0 / 0	0	4 / 8	0.0013	0.055
6 h	14	103	0.141	2.95×10^{-4}	0 / 0	0	6 / 12	0.0038	0.061
20 h	13	844	0.141	6.52×10^{-4}	0 / 0	25	7 / 11	0.0094	0.130

The trail calibration chose length zero on every exposure, which the expectation registered in PLAN.md did not predict. The reason is in the result file: the release blinds its hits, so a whole exposure holds 0, 70, 103, 844 trigger pixels. At the longest exposure the test does see the excess on the right side — 13 single electrons within 5 superpixels downstream of a trigger against 2 upstream, $p = 0.0037$ — but the corrected threshold is $p < 6.3e-05$. Part of that strictness is an artefact: the correction counts 80 distance blocks per direction, and on an image 16 rows tall only three of them can ever be filled. Correcting for the tests that can exist would ask for about 1.7×10^{-3} , which this p-value still does not meet, so the decision stands. The halo radius behaves the same way: 0 where there are no triggers, 25 column widths at the longest exposure.

What the checks caught

Every error below was found by a test or a diagnostic against the simulator's truth, fixed, and recorded.

- The pixel threshold of every “adaptive” mask was set from the simulator's true mean charge: truth leaking into the masks, and at the surface a threshold of 0.50 e that turned read noise into fake events. Found by an independent review; the threshold is now measured from each image's charge histogram and every result was re-run.
- The simulator redrew hot columns in every image, which would have invalidated every calibration on a stack.
- The hot-column mask flagged the innocent neighbours of each hot column (17 of them in a test).
- The halo test treated its reference rate as exact; a 4σ fluctuation gave a halo where there was none.

- At the surface no pixel is far from every muon track: the first halo calibration fell back to its largest radius and masked the whole image. It now compares each annulus with everything outside it.
- Crossing muons merged into wide clusters that the track criterion rejected (24 % of muon pixels missed); tracks cut by the image border were missed too.
- Hot columns were calibrated before charge-transfer trails, whose vertical trails from muons made 69 of 81 flagged columns false.
- In one seed the halo radius, chosen by significance alone, reached 125 px on the surface sensor and masked 99.9 % of the image. The radius now maximises a figure of merit estimated from the data within the significant range; all seeds were re-run, plus two seeds never used during development.

Limitations

- Defect populations (hot columns, trail lengths, serial hits, clusters) are scenario values, not measurements.
- One figure of merit; the oracle is tuned mask by mask, not jointly, so a combined adaptive mask can exceed it.
- Adaptive trail lengths are limited by the number of bright pixels in the calibration stack: they are short when data are few.
- The muon figure of merit counts pixels, which weighs every missed track pixel heavily.
- The adaptive muon mask uses the sensor's diffusion model and the minimum-ionising charge per length, and the simulated muons carry exactly that mean charge with no Landau tail, so its charge tolerance is tested against the model it was built from.
- The defects are switched on one at a time, which finds a mask firing on another single defect, but not what a mask does when several defects overlap in the same pixels.
- The oracle is the best fixed mask on a grid of parameters. Every seed passed a check that no optimum lies on an edge of its grid other than a physical bound (a length of zero, a factor of one), but the grid is still finite.
- An adaptive mask cannot tell injected signal from dark current, so it estimates signal as every uniform single electron; the evaluation counts only the injected signal. At the surface, where dark current is ~20 times the signal, the adaptive halo therefore chooses not to mask and reaches ~0.7 of the oracle. The figure of merit was not changed after this was seen (the no-mask reference and wider grids were added later, for other reasons).
- On the real release only four of the six masks can be checked at all: it publishes no counterpart to the low-energy-cluster mask, and its binned superpixels make the muon mask's geometry meaningless, so that mask was not run there.

Reproduce

```
uv sync --group dev
uv run python scripts/fetch_public_data.py
uv run pytest
uv run python analysis/reproduce_release_rate.py
uv run python analysis/check_release_mask_bits.py
uv run python analysis/adaptive_masks_on_public_data.py
uv run python analysis/null_false_positive_rates.py 50
uv run python analysis/muon_mask_null_rate.py 100
uv run python analysis/cross_defect_false_positives.py 20
for seed in 20260915 20260916 20260917 20260920 20260921; do
    uv run python analysis/compare_masks_across_sensors.py 4 --seed $seed --out results/compare_masks/seed_$seed
done
uv run python analysis/figure_release_rate.py
uv run python analysis/figure_null_rates.py
uv run python analysis/figure_cross_defect.py
uv run python analysis/figure_compare_masks.py
uv run python analysis/build_report.py
uv run python scripts/make_pdf.py
```

Every output has a `.provenance.json` sidecar with the script, git commit, input hashes, parameters and seed; [PROVENANCE.md](#) says how each result was checked.

What that reproduction covered, as of 2026-09-16: those commands were run in a fresh clone of the frozen commit, in the pinned environment, and the results compared with the committed files at a relative tolerance of $1e-9$. Every result came out identical, largest relative difference zero: R1, R2, R3 and all five R4 seeds, the summary the figure script derives from them, and the figure itself byte for byte. This sentence is written by hand, unlike every number above it, because it describes a run made outside this repository.